

Learning Matryoshka Contrastive Node Embeddings for Resource-Adaptive Deployment

Anonymous submission

Abstract

Graph contrastive learning (GCL) has emerged as a powerful paradigm for learning node embeddings without task-specific supervision. However, existing methods typically learn embeddings with a fixed dimensionality, limiting their applicability to deployment environments with different memory, latency, and computation budgets. Training separate models for different embedding dimensions is possible but incurs substantial training and storage costs. Inspired by Matryoshka Representation Learning, we investigate whether a single GCL model can learn nested node embeddings whose prefixes can be directly truncated and deployed at multiple dimensions. This objective is non-trivial for graphs because node embeddings are primarily defined by the relational structures they preserve. Moreover, higher-dimensional prefixes should not simply reproduce lower-dimensional ones, but should progressively resolve node pairs that remain difficult at smaller dimensions. To this end, we propose **Matryoshka Contrastive Node Embeddings (MCNE)**, a resource-adaptive GCL framework that jointly learns multiple prefix embeddings using a single GNN encoder. MCNE introduces three dimension-aware components: Cross-Dimensional Mutual Distillation, which aligns each lower-dimensional prefix with the highest-dimensional prefix; Hard-Pair Evolutionary Mining, which guides larger prefixes to distinguish difficult node pairs left unresolved by smaller prefixes; and Dimension-Adaptive Loss Scheduling, which balances contrastive learning across embedding capacities. Experiments on three large-scale graph datasets show that MCNE learns a single nested embedding space that supports efficient deployment under different resource budgets. Across embedding dimensions, it achieves competitive or superior performance while avoiding independent training and post-hoc compression. Our code can be found at the "Code and Data Supplement" on OpenReview.

1 Introduction

Graph-structured data are ubiquitous in real-world applications, ranging from social networks and recommender systems to fraud detection and traffic systems (Ju et al. 2024). A common way to exploit such data is to learn node embeddings, which map nodes into continuous vector spaces and support a wide spectrum of downstream tasks, such

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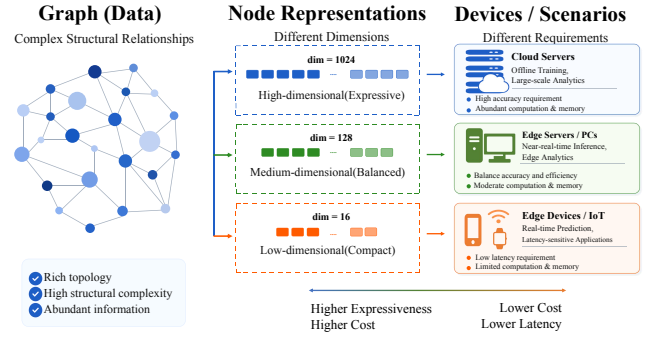


Figure 1: Diverse embedding dimensions for a graph.

as node classification, link prediction, and clustering (Kipf and Welling 2017; Zhang 2022; Liu et al. 2026). Recently, graph contrastive learning (GCL) has become a promising paradigm for learning node embeddings without task-specific labels (Zhu et al. 2021; Liu et al. 2022; Liang et al. 2025). Representative methods construct multiple augmented views of a graph and train a graph neural network (GNN) encoder to maximize the agreement between representations of the same node while contrasting them against other nodes.

Despite their success, existing GCL methods usually assume a fixed embedding dimension during training and deployment. This assumption is increasingly restrictive in practical graph applications. As shown in Figure 1, node embeddings are often served under heterophilous resource constraints in real-world deployment. For example, cloud servers may afford high-dimensional embeddings to maximize accuracy, while edge devices, mobile terminals, or latency-sensitive retrieval systems may only support compact embeddings due to memory, bandwidth, and computation limitations (Ma et al. 2025; Yin et al. 2025). Consequently, different scenarios often require different embedding dimensions. A straightforward solution is to train separate graph encoders or projection heads for each target dimension. However, this significantly increases training cost, storage overhead, and maintenance complexity. Alternatively, one may train a high-dimensional model and compress it into smaller models by post-hoc dimensionality reduction or knowledge distillation (Hinton, Vinyals, and Dean 2015), but such pro-

cedures require additional training and may not preserve the contrastive structure learned by the original model.

This raises a direct question: *Can a single graph contrastive model produce node embeddings that can be cropped to different dimensions and directly deployed under different resource budgets?* We refer to this setting as *resource-adaptive deployment*. Instead of training or compressing a separate model for each embedding dimension, the goal is to learn one nested embedding space in which every prefix embedding is immediately usable and remains discriminative for downstream graph tasks.

A natural inspiration comes from Matryoshka Representation Learning (MRL), which trains nested representations so that smaller prefixes can be directly used for efficient inference (Kusupati et al. 2022). However, directly transferring this idea to GCL is non-trivial. First, in GCL, the semantics of node embeddings are primarily reflected by the relational structure they induce over nodes, rather than by each embedding vector independently. Therefore, enforcing consistency across different-dimensional prefixes in raw vector space is inappropriate: since a lower-dimensional prefix is literally a sub-vector of a higher-dimensional embedding, naive vector-level alignment may suppress the information carried by additional dimensions. Second, GCL relies heavily on positive-negative discrimination. Higher-dimensional prefixes should not simply imitate lower-dimensional ones; instead, the extra dimensions should be encouraged to resolve the hard node pairs that smaller prefixes fail to distinguish.

To address these challenges, we propose Matryoshka Contrastive Node Embeddings (MCNE), a simple yet effective framework for learning nested node embeddings under GCL. MCNE uses a GNN encoder to produce a full-dimensional node embedding matrix, from which multiple prefix embeddings are cropped according to pre-defined target dimensions. These prefixes share parameters and are optimized jointly within a single contrastive training process, enabling one trained model to support multiple deployment budgets. To make different prefixes mutually beneficial, MCNE introduces *Cross-Dimensional Mutual Distillation*, which transfers relational knowledge between each lower-dimensional prefix and the highest-dimensional prefix by aligning their induced relational structures rather than their raw embedding vectors. To ensure that additional dimensions contribute new discriminative information, MCNE further designs *Hard-Pair Evolutionary Mining*, which encourages larger prefixes to focus on hard node pairs that smaller prefixes fail to separate. Finally, MCNE adopts *Dimension-Adaptive Loss Scheduling* to balance the training signals across prefixes with different capacities and contrastive difficulties. In this way, MCNE extends the idea of MRL to GCL while respecting the relational and contrastive nature of embeddings.

We evaluate MCNE on three large-scale graphs under multiple deployment dimensions. Experimental results show that MCNE learns a single nested embedding space whose cropped prefixes can be directly deployed with strong performance. In summary, our contributions are as follows:

- The problem of learning Matryoshka contrastive node embeddings is introduced for resource-adaptive graph deployment, allowing node embeddings of multiple tar-

get dimensions to be directly cropped from a single full-dimensional representation.

- A graph contrastive learning framework, MCNE, is proposed to jointly optimize nested prefix embeddings with shared GNN parameters and support efficient multi-dimensional deployment without additional training.
- Comprehensive experiments on large-scale graphs demonstrate that MCNE achieves strong performance across embedding dimensions and provides practical benefits for resource-adaptive applications.

2 Related Work

2.1 Graph Contrastive Learning

Graph Contrastive Learning (GCL) learns graph representations by maximizing the agreement between semantically related views without task-specific labels (Liu et al. 2022). Early methods maximize mutual information between local and global representations or contrast the same nodes across augmented graph views (Veličković et al. 2019; Zhu et al. 2020, 2021). Other approaches eliminate explicit negative samples through feature decorrelation (Zhang et al. 2021) or bootstrapping (Thakoor et al. 2022). Recent studies have explored unified graph augmentations and representation scattering (Zhuo et al. 2024; He et al. 2024), progressive and spectral view construction (Zhao et al. 2025; Chen, Lei, and Wei 2024), and calibration of noisy contrastive supervision (Lin et al. 2025). Despite these advances, existing methods generally learn embeddings with a fixed dimension, whereas our work enables a single GCL model to support multiple deployment dimensions.

2.2 Matryoshka Representation Learning

Matryoshka Representation Learning (MRL) learns nested coarse-to-fine representations by jointly optimizing multiple prefixes of a single embedding (Kusupati et al. 2022). Each prefix can be used independently at inference time, allowing the embedding dimension to be adjusted according to computational and storage budgets without training separate models or applying post-hoc compression. This principle has been extended to sentence representation learning (Li et al. 2025), image generation (Gu et al. 2024), multimodal representation learning (Hu et al. 2024; Cai et al. 2025), and information retrieval (Zhang et al. 2025). However, existing MRL methods mainly focus on independent samples or task-specific supervision and have not explored nested node embeddings. Our work extends MRL to graphs by jointly learning prefix embeddings while considering the relational and contrastive characteristics of node representations.

3 Preliminaries

3.1 Problem Definition

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ be an attributed graph, where $\mathcal{V}$ is the set of N nodes, $\mathcal{E}$ is the edge set, and $\mathbf{X} \in \mathbb{R}^{N \times d_x}$ is the node feature matrix. A GNN encoder f_θ maps $\mathcal{G}$ to a full-dimensional embedding matrix $\mathbf{H} = f_\theta(\mathcal{G}) \in \mathbb{R}^{N \times d_{\max}}$. Given a pre-defined set of target dimensions $\mathcal{D} = \{d_1, d_2, \dots, d_K\}$ with

$d_1 < d_2 < \dots < d_K = d_{\max}$, the i -th *prefix embedding* is obtained by truncating the leading dimensions:

$$\mathbf{H}^{(i)} = \mathbf{H}[:, :d_i] \in \mathbb{R}^{N \times d_i}, \quad i \in \{1, \dots, K\}. \quad (1)$$

Our goal is to learn a single encoder f_θ such that for every $d_i \in \mathcal{D}$, the prefix embedding $\mathbf{H}^{(i)}$ is a high-quality d_i -dimensional node representation that can be directly deployed without additional training or post-hoc compression. We refer to this setting as *resource-adaptive deployment*: a single trained model accommodates heterophilous deployment scenarios by simply cropping the embedding to the required dimension.

3.2 Graph Contrastive Learning

Graph contrastive learning (GCL) provides the backbone training paradigm for our framework. Typically, GCL generates two augmented views $\mathcal{G}_1$ and $\mathcal{G}_2$ from $\mathcal{G}$ via stochastic augmentation strategies such as feature masking and edge perturbation. The encoder f_θ maps each view to a node embedding matrix $\mathbf{H}_r = f_\theta(\mathcal{G}_r) \in \mathbb{R}^{N \times d_{\max}}$, $r \in \{1, 2\}$, where $\mathbf{h}_{r,u}$ denotes the embedding of node u in view r . Representations of the same node across views form positive pairs, while those of different nodes form negatives. The dominant objective is InfoNCE (Chen et al. 2020):

$$\mathcal{L}_{\text{InfoNCE}} = - \sum_{u=1}^N \log \frac{\exp(\text{sim}(\mathbf{h}_{1,u}, \mathbf{h}_{2,u})/\tau)}{\sum_{v=1}^N \exp(\text{sim}(\mathbf{h}_{1,u}, \mathbf{h}_{2,v})/\tau)}, \quad (2)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity and τ is the temperature. We write $\mathcal{L}_{\text{InfoNCE}}^{(i)}$ for this loss evaluated on the i -th prefix embeddings $\mathbf{H}_r^{(i)} = \mathbf{H}_r[:, :d_i]$, with $\mathbf{h}_{r,u}$ replaced by $\mathbf{h}_{r,u}^{(i)}$. Existing GCL methods train and deploy embeddings at a single fixed dimension; our work extends this paradigm to nested prefix embeddings.

4 Method

4.1 Overall Framework

The overview of our full framework is shown in Figure 2. Following MRL (Kusupati et al. 2022), our goal is to learn a single GNN encoder f_θ that produces a full-dimensional embedding $\mathbf{H}_r = f_\theta(\mathcal{G}_r) \in \mathbb{R}^{N \times d_{\max}}$ for each augmented view $\mathcal{G}_r$, such that every prefix $\mathbf{H}_r^{(i)} = \mathbf{H}_r[:, :d_i]$ is independently usable at inference. A natural baseline applies InfoNCE (Eq. 2) at each target dimension:

$$\mathcal{L}_{\text{MRL}} = \frac{1}{K} \sum_{i=1}^K \mathcal{L}_{\text{InfoNCE}}^{(i)}, \quad (3)$$

where $\mathcal{L}_{\text{InfoNCE}}^{(i)}$ is the InfoNCE loss on prefix embeddings $\mathbf{H}_r^{(i)}$. At inference, any prefix $\mathbf{H}^{(i)} = f_\theta(\mathcal{G})[:, :d_i]$ is directly deployed. However, this direct extension has three shortcomings. (i) *Semantic inconsistency*: without cross-dimensional interaction, different prefixes may develop conflicting relational structures, undermining the coherence of the nested embedding space. (ii) *Redundant learning*: training every prefix on equally-weighted pairs encourages

higher-dimensional prefixes to re-learn what smaller ones already capture, wasting their additional capacity. (iii) *Unbalanced optimization*: equal loss weights ignore that larger prefixes have greater representational capacity and should bear proportionally more optimization responsibility. MCNE addresses these with three complementary components: *Cross-Dimensional Mutual Distillation* (CDMD), *Hard-Pair Evolutionary Mining* (HPEM), and *Dimension-Adaptive Loss Scheduling* (DALs).

4.2 Cross-Dimensional Mutual Distillation

When prefix dimensions are optimized independently, there is no mechanism to coordinate the relational structures they induce. The same node pair may be treated as similar under one prefix but dissimilar under another, violating the coherence a nested embedding space requires. While embeddings at different dimensions cannot be compared directly in Euclidean space, the pairwise similarity matrices they induce encode the same underlying graph semantics. We therefore enforce cross-dimensional consistency at the *affinity* level rather than the embedding level.

Relational affinity vectors. For each dimension d_i and a mini-batch $\mathcal{B}$ of size B , we compute the cross-view pairwise similarity matrix:

$$\mathbf{S}_{u,v}^{(i)} = \text{sim}(\mathbf{h}_{1,u}^{(i)}, \mathbf{h}_{2,v}^{(i)}), \quad \mathbf{S}^{(i)} \in \mathbb{R}^{B \times B}, \quad (4)$$

where diagonal entries are positive pairs and off-diagonal entries are negatives. Since cosine similarities lie in $[-1, 1]$, we apply ReLU to discard negative affinities and obtain a non-negative relational affinity vector:

$$\mathbf{q}^{(i)} = \text{ReLU}(\text{vec}(\mathbf{S}^{(i)})) + \epsilon, \quad (5)$$

where $\epsilon > 0$ ensures non-zero entries for numerical stability.

Distillation objective. The highest-dimensional prefix carries the richest relational information and serves as the global semantic reference. We align every lower-dimensional prefix to it through bidirectional KL divergence on the affinity vectors:

$$\mathcal{L}_{\text{CDMD}}^{(i)} = \frac{1}{2B} \left[D_{\text{KL}}(\mathbf{q}^{(i)} \parallel \text{sg}(\mathbf{q}^{(K)})) + D_{\text{KL}}(\mathbf{q}^{(K)} \parallel \text{sg}(\mathbf{q}^{(i)})) \right], \quad (6)$$

where $\text{sg}(\cdot)$ is the stop-gradient operator. The forward direction transfers fine-grained relational structure from the richest prefix to smaller ones; the reverse direction regularizes the largest prefix with stable coarse-grained semantics from smaller ones. Stop-gradient prevents both affinity vectors from drifting toward each other simultaneously. The full CDMD loss averages over lower-dimensional prefixes:

$$\mathcal{L}_{\text{CDMD}} = \frac{1}{K-1} \sum_{i=1}^{K-1} \mathcal{L}_{\text{CDMD}}^{(i)}. \quad (7)$$

CDMD ensures all prefixes agree on which node pairs are similar. Yet relational alignment alone does not determine how the additional capacity of higher-dimensional prefixes should be spent, and it does not identify which node pairs remain hard and deserve stronger supervision.

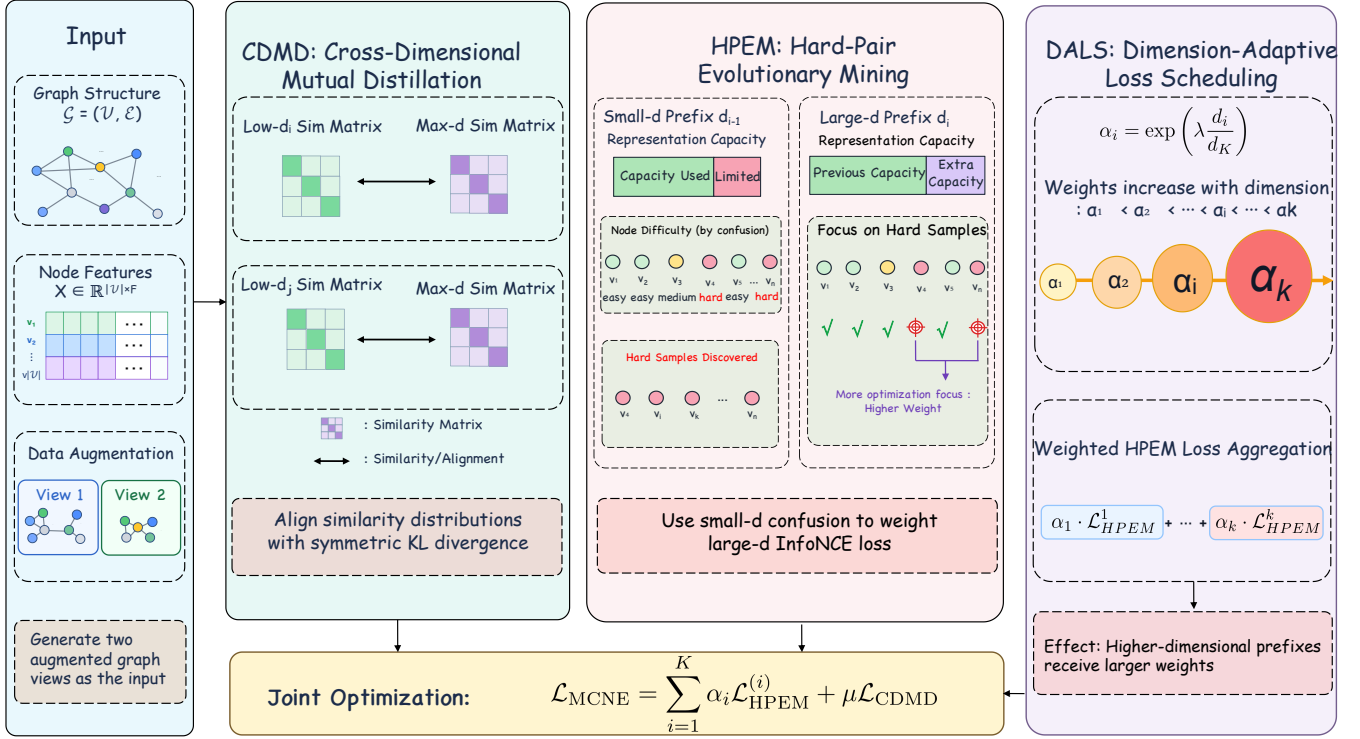


Figure 2: Overview of MCNE. A shared GNN encoder processes two augmented graph views and produces a full-dimensional embedding matrix, cropped into nested prefix embeddings at pre-defined target dimensions. CDMD aligns relational affinities across prefixes, HPEM steers larger prefixes toward node pairs left unresolved by smaller ones, and DALs assigns capacity-proportional loss weights.

4.3 Hard-Pair Evolutionary Mining

Even with relational consistency enforced by CDMD, each prefix is still trained on the same contrastive pairs with uniform weights. The extra $d_i - d_{i-1}$ dimensions introduced at prefix i should not merely duplicate information already captured by prefix $i-1$; rather, they should be directed toward the node pairs that prefix $i-1$ failed to separate. A lower-dimensional prefix naturally reveals its own failure modes: a small or negative positive-to-hardest-negative margin signals that the local neighborhood of that anchor is unresolved and deserves stronger supervision at the next dimension.

Confusion margin. For each anchor u , we measure its difficulty at dimension d_{i-1} via:

$$c_u^{(i-1)} = \text{sim}(\mathbf{h}_{1,u}^{(i-1)}, \mathbf{h}_{2,u}^{(i-1)}) - \max_{v \neq u} \text{sim}(\mathbf{h}_{1,u}^{(i-1)}, \mathbf{h}_{2,v}^{(i-1)}). \quad (8)$$

A large positive margin indicates an easy anchor; a small or negative margin indicates one whose neighborhood prefix $i-1$ cannot adequately resolve.

Rewighted objective. We convert confusion margins into per-anchor weights via a softmax that concentrates mass on harder cases:

$$w_u^{(i)} = \frac{\exp(-\beta \text{sg}(c_u^{(i-1)}) / \tau_i)}{\sum_{a \in \mathcal{B}} \exp(-\beta \text{sg}(c_a^{(i-1)}) / \tau_i)}, \quad (9)$$

where $\beta = \text{softplus}(\hat{\beta}) > 0$ is a learnable scaling parameter, $\text{sg}(\cdot)$ blocks gradients to the preceding prefix, and τ_i is the dimension-adaptive temperature defined in Section 4.4. For $i = 1$, we set $w_u^{(1)} = 1/B$ (uniform). The HPEM loss at dimension d_i is:

$$\mathcal{L}_{\text{HPEM}}^{(i)} = \sum_{u \in \mathcal{B}} w_u^{(i)} \mathcal{L}_{\text{InfoNCE},u}^{(i)}, \quad (10)$$

where $\mathcal{L}_{\text{InfoNCE},u}^{(i)}$ is the per-node InfoNCE loss at prefix i with temperature τ_i . For $i \geq 2$, anchors hardest for the preceding prefix receive proportionally more weight, so each successive prefix is steered toward the ambiguities its predecessor left unresolved.

4.4 Dimension-Adaptive Loss Scheduling

HPEM determines *where* each prefix should direct its capacity, but two further questions arise: how sharp should the contrastive objective be at each dimension, and how much should each prefix contribute to the total loss? Both questions are rooted in the same geometric fact: for random unit vectors in $\mathbb{R}^d$, the variance of cosine similarity scales as $O(1/d)$, so similarity values concentrate more tightly around zero as dimensionality grows (Cai, Fan, and Jiang 2013). A fixed temperature τ therefore induces different effective contrastive difficulties across prefix dimensions: lower-dimensional pre-

fixes face a less sharp loss landscape and receive weaker optimization signal, while higher-dimensional prefixes occupy the opposite regime. DALs addresses both issues through a dimension-adaptive temperature and a capacity-proportional loss weighting.

Dimension-adaptive temperature. We parameterize the temperature for prefix i as:

$$\tau_i = \tau_0 \exp\left(\phi_1 \frac{d_i}{d_K} + \phi_2\right), \quad (11)$$

where τ_0 is the base temperature and ϕ_1, ϕ_2 are learnable scalars initialized to zero so that $\tau_K \approx \tau_0$ at the start of training. This temperature is shared by both the confusion-to-weight mapping in HPEM (Eq. 9) and the per-node InfoNCE loss at dimension d_i , allowing the model to jointly calibrate contrastive sharpness and sample focus across dimensions.

Capacity-proportional loss weights. Beyond temperature, different prefix dimensions play asymmetric roles during optimization: low-dimensional prefixes rely more on CDMD to receive structural guidance from larger prefixes, while high-dimensional prefixes should take on greater independent optimization responsibility. We reflect this asymmetry by assigning each HPEM objective a dimension-dependent weight:

$$\alpha_i = \exp\left(\lambda \frac{d_i}{d_K}\right), \quad (12)$$

where λ is a learnable scalar initialized to zero, yielding uniform weights at the start of training. As λ grows positive, higher-dimensional objectives exert proportionally greater influence on optimization without suppressing gradient flow to smaller prefixes. DALs introduces no additional loss term; it modulates the relative importance of HPEM objectives through τ_i and $\{\alpha_i\}$.

Full objective. The full MCNE objective combines all three components:

$$\mathcal{L}_{\text{MCNE}} = \sum_{i=1}^K \alpha_i \mathcal{L}_{\text{HPEM}}^{(i)} + \mu \mathcal{L}_{\text{CDMD}}, \quad (13)$$

where $\mu > 0$ controls the strength of cross-dimensional distillation and $\{\alpha_i\}$ are the DALs weights.

5 Experiments

5.1 Experiment Setup

Datasets. We evaluate our proposed method on three large-scale graph datasets from the Open Graph Benchmark (OGB (Hu et al. 2020)), with sizes ranging from approximately 169K to 2.4M nodes. These datasets cover diverse domains and graph types: two homophilous graphs, OGBN-ARXIV, a citation network, and OGBN-PRODUCTS, a product co-purchasing network; and one heterophilous academic graph, OGBN-MAG. Detailed dataset statistics and descriptions are provided in the Appendix B.

Baselines. We compare our model against representative GCL methods, including DGI (Veličković et al. 2019), GraphCL (You et al. 2020), GRACE (Zhu et al. 2020), CCA-SSG (Zhang et al. 2021), SSGE (Liu et al. 2025), and lrGAE (Li et al. 2026). In addition, we incorporate MRL (Kusupati et al. 2022) into GCL as a resource-adaptive baseline. Further details are provided in the Appendix C.

Implementation Details. Following the typical GCL pipeline, we evaluate node classification performance using a linear classifier trained on frozen graph representations. We evaluate all methods at seven target embedding dimensions, $\mathcal{D} = \{32, 64, 128, 256, 384, 512, 768\}$, ranging from compact 32-dimensional embeddings to the full 768-dimensional representation. All results are reported as the mean and standard deviation over five independent runs using random seeds 0, 1, 2, 3, and 4. All methods are implemented in PyTorch, and the experiments are conducted on a machine equipped with an NVIDIA Tesla V100-SXM2 GPU with 32 GB of memory. For MCNE, we first optimize the standard MRL objective during a warm-up stage and then continue training with the proposed MCNE objective. Detailed hyperparameter settings are provided in the Appendix D.

5.2 Performance Comparison

Overall performance. Table 1 reports the node classification accuracy on three datasets across seven embedding dimensions. MCNE ranks first in 19 of the 21 dataset–dimension settings and second in the remaining two. Among the next-best results in the 19 settings where MCNE ranks first, 18 are produced by independently trained models, which must repeat the complete representation training, embedding inference, and downstream classification workflow for all seven dimensions. In contrast, MCNE trains the encoder and infers the full-dimensional embeddings only once, followed by separate linear evaluation of the seven prefixes. These results show that MCNE not only improves predictive performance across embedding dimensions but also provides substantial gains in overall training efficiency. The following analysis examines its performance separately in the low- and higher-dimensional regimes.

Performance in the Low-Dimensional Regime (32–128).

For devices relying on low-dimensional embeddings, maintaining usable predictive accuracy is essential. At 32 dimensions, standard MRL underperforms independently trained models by 8.74, 4.14, and 10.73 percentage points on OGBN-ARXIV, OGBN-MAG, and OGBN-PRODUCTS, respectively. In contrast, MCNE raises the corresponding accuracies to 65.56%, 30.77%, and 69.26%, surpassing the strongest independently trained models by 0.78, 0.55, and 2.36 percentage points. Moreover, MCNE ranks first in all nine settings from 32 to 128 dimensions, demonstrating that it improves compact prefixes to a practically usable level without sacrificing performance relative to independently trained models.

Performance in the Higher-Dimensional Regime (256–768).

For deployments using high-dimensional embeddings, achieving the highest possible predictive accuracy is the primary objective, and even modest improvements can

OGBN-ARXIV (169K nodes, 1.17M edges)							
Method	32	64	128	256	384	512	768
DGI	58.24±0.36	63.23±0.28	67.86±0.26	69.59±0.22	69.60±0.10	70.01±0.37	70.12±0.30
GraphCL	64.64±0.30	67.54±0.10	68.71±0.39	68.37±0.22	68.43±0.67	68.28±0.45	68.65±0.28
GRACE	64.78±0.10	67.26±0.00	67.73±0.14	67.66±0.14	67.61±0.20	68.16±0.14	68.56±0.14
CCA-SSG	58.17±0.26	64.33±0.20	67.59±0.10	69.17±0.24	69.80±0.30	70.03±0.44	70.41±0.26
SSGE	61.42±0.22	66.74±0.28	68.43±0.30	69.52±0.17	70.07±0.30	70.25±0.30	70.64±0.20
lrGAE	50.74±0.24	59.01±0.24	63.17±0.38	65.05±0.51	65.79±0.20	65.65±0.17	66.42±0.17
MRL	56.04±0.42	62.14±0.37	66.31±0.17	68.32±0.14	68.43±0.20	68.83±0.10	68.96±0.14
MCNE	65.56±0.17	68.32±0.14	69.63±0.17	70.48±0.24	70.52±0.33	70.68±0.26	70.70±0.36

OGBN-MAG (736K nodes, 10.79M edges)							
Method	32	64	128	256	384	512	768
DGI	28.34±0.44	30.13±0.55	30.07±0.17	30.55±0.32	29.97±0.64	30.83±0.41	28.98±0.10
GraphCL	28.89±0.53	29.33±0.45	28.98±0.47	28.12±0.34	28.22±0.28	28.12±0.40	27.75±0.37
GRACE	30.22±0.17	32.07±0.20	32.48±0.10	32.32±0.17	32.30±0.22	32.56±0.20	32.71±0.17
CCA-SSG	25.77±0.22	27.81±0.32	29.24±0.36	30.49±0.37	31.11±0.33	31.30±0.20	31.51±0.14
SSGE	27.54±0.10	29.33±0.00	30.87±0.10	33.01±0.24	33.73±0.30	33.57±0.14	34.18±0.17
lrGAE	24.73±0.64	27.92±0.22	30.17±0.57	31.11±0.20	31.56±0.92	32.11±0.73	30.12±0.37
MRL	26.08±0.10	29.85±0.20	32.40±0.32	33.17±0.10	33.22±0.24	33.00±0.26	33.23±0.10
MCNE	30.77±0.28	32.68±0.14	33.79±0.53	33.79±0.22	34.05±0.79	33.94±0.48	33.69±0.84

OGBN-PRODUCTS (2.45M nodes, 61.86M edges)							
Method	32	64	128	256	384	512	768
DGI	63.59±0.17	67.74±0.20	70.64±0.20	73.66±0.17	74.00±0.24	74.38±0.65	72.65±0.95
GraphCL	65.29±1.27	67.96±0.91	70.00±0.92	72.55±0.63	73.78±0.58	73.81±0.28	72.61±0.95
GRACE	66.90±0.14	71.82±0.10	74.57±0.17	75.70±0.44	76.09±0.40	76.61±0.60	76.74±0.39
CCA-SSG	59.37±0.10	62.61±0.14	65.99±0.14	69.18±0.20	71.57±0.26	72.75±0.44	74.28±0.85
SSGE	64.25±0.10	68.26±0.22	70.59±0.26	72.40±0.33	71.46±1.68	73.13±1.07	72.68±1.02
lrGAE	55.42±0.20	61.47±0.26	65.91±0.14	69.18±1.26	70.20±0.84	70.34±1.18	71.64±1.22
MRL	56.17±0.14	64.13±0.14	72.22±0.14	74.27±0.10	74.37±0.37	74.56±0.14	74.72±0.20
MCNE	69.26±0.14	73.89±0.17	76.42±0.14	76.46±0.76	76.43±0.20	77.18±0.24	76.45±0.63

Table 1: Node classification accuracy (%) of different methods on OGBN-ARXIV, OGBN-MAG, and OGBN-PRODUCTS under different embedding dimensions. All results are averaged over five independent runs and reported as mean±std.

be important. MCNE ranks first in ten of the twelve settings from 256 to 768 dimensions. On OGBN-ARXIV, it consistently outperforms the strongest competing models by 0.89, 0.45, 0.43, and 0.06 percentage points at 256, 384, 512, and 768 dimensions, respectively. It also improves the best accuracy by 0.37 points at 512 dimensions on OGBN-MAG and by 0.57 points on OGBN-PRODUCTS. In the remaining two settings at 768 dimensions, MCNE ranks second and trails the best independently trained models by only 0.49 and 0.29 percentage points. These results show that MCNE preserves the high accuracy required by resource-rich deployments while supporting all target dimensions with a single model.

5.3 Efficiency Comparison

Table 2 compares the wall-clock time required to obtain representations at all seven target dimensions. Under the same training settings within each dataset, MCNE achieves

Dataset	GRACE (7 runs)	MRL (1 run)	MCNE (1 run)
OGBN-ARXIV	34:37:24	06:12:44	06:27:43
OGBN-MAG	16:48:32	04:26:37	05:12:48
OGBN-PRODUCTS	66:35:58	10:10:52	10:26:29

Table 2: Wall-clock training time for GRACE, MRL and MCNE on three datasets.

a $3.22\times$ – $6.38\times$ speedup over independently training seven GRACE models because a single model produces all seven nested representations. Compared with standard MRL, the additional training overhead remains moderate and arises from cross-dimensional distillation, hard-pair mining, and dimension-adaptive optimization. Thus, MCNE largely pre-

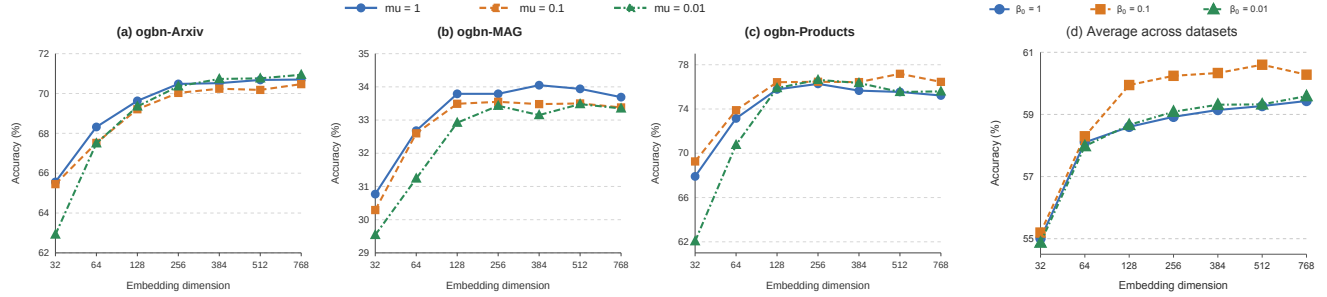


Figure 3: Sensitivity analysis of the loss weight μ and β_0 across seven embedding dimensions.

Method	32	64	128	256	384	512	768
MRL	46.10	52.04	56.98	58.59	58.67	58.80	58.97
+ CDMD	54.03	57.42	59.46	60.02	60.20	59.86	59.43
+ HPEM	51.33	54.96	58.45	58.89	59.10	59.03	59.06
+ HPEM + DALs	52.74	56.55	58.73	59.59	59.39	59.47	59.22
MCNE	55.20	58.30	59.95	60.24	60.33	60.60	60.28

Table 3: Ablation study in terms of averaged node classification accuracy (%) across three datasets.

serves the efficiency of single-model Matryoshka training while producing substantially stronger representations. Although OGBN-MAG is larger than OGBN-ARXIV, it requires fewer epochs to converge and therefore has a shorter overall training time. Since graph scales and training configurations differ across datasets, efficiency is compared in each dataset.

5.4 Ablation Study

To assess the contribution of each proposed component, we conduct an ablation study by selectively incorporating CDMD, HPEM, and DALs into the standard MRL baseline. For each embedding dimension, node classification accuracy is averaged across the three datasets. As shown in Table 3, introducing either CDMD or HPEM consistently improves upon MRL across all embedding dimensions, demonstrating the effectiveness of both components. CDMD generally provides larger individual gains, whereas HPEM also yields clear improvements, particularly for compact representations. Applying DALs to HPEM further increases accuracy at every dimension by 0.16–1.59 percentage points, confirming that adaptively balancing dimension-specific HPEM losses leads to more effective optimization. The benefits of the proposed components are especially pronounced at lower dimensions, indicating that they alleviate the information bottleneck caused by limited embedding capacity and enhance the discriminative power of compact representations. When CDMD and HPEM with DALs are combined, the complete MCNE achieves the best result at every dimension, outperforming MRL by 1.31–9.10 percentage points. Although the improvements gradually diminish at higher dimensions as the baseline approaches saturation, the consistent superiority of MCNE demonstrates that CDMD and HPEM with DALs pro-

vide complementary supervision and deliver stable benefits across a broad range of embedding dimensions.

5.5 Sensitivity Analysis

We first investigate the sensitivity of MCNE to the loss weight μ by varying it within $\{0.01, 0.1, 1.0\}$ while keeping all other hyperparameters unchanged. As shown in Figure 3(a)–(c), the optimal value varies across datasets: $\mu = 1.0$ generally performs better on OGBN-ARXIV and OGBN-MAG, whereas $\mu = 0.1$ achieves better overall performance on OGBN-PRODUCTS. The influence of μ is more pronounced at lower dimensions, particularly at 32 dimensions, while the performance differences generally decrease as the embedding dimension increases. This indicates that low-dimensional prefixes are more sensitive to the balance among training objectives, whereas MCNE remains relatively robust to μ at medium and high dimensions.

We further analyze the sensitivity of MCNE to the initial value β_0 of the learnable scaling parameter β , considering $\beta_0 \in \{0.01, 0.1, 1.0\}$. At each embedding dimension, the accuracy under each initialization is averaged across the three datasets. As shown in Figure 3(d), $\beta_0 = 0.1$ achieves the highest average accuracy across all embedding dimensions. The differences among the initializations are relatively small at 32 and 64 dimensions but become more evident from 128 dimensions onward, where $\beta_0 = 0.1$ outperforms the stronger alternative by approximately 0.69–1.29 percentage points. These results show that MCNE remains stable under different β_0 , with $\beta_0 = 0.1$ providing the most reliable overall performance across datasets and embedding dimensions.

6 Conclusion

We presented MCNE, a unified framework for learning nested graph representations with a single model. By integrating CDMD, HPEM, and DALs, MCNE learns prefix embeddings that can be directly cropped for different deployment budgets without retraining the representation model or applying post-hoc compression. Comprehensive experiments on three large-scale graphs demonstrate strong performance across dimensions, particularly for compact embeddings, while avoiding the cost of training separate models.

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